

Calculating Artifact Roll Outcomes

Brandon Fowler

July 12, 2026

This document provides an overview of the approach taken to calculate all possible outcomes for an artifact in Genshin Impact as used by Artifact Odds.

Contents

1	Sub-Stat Probability	2
2	Roll Probability	2
2.1	Total Number of Roll Possibilities	3
2.2	Roll Permutations	3
2.3	Initial and Roll Values	3
2.4	Total Probability	4
2.5	Generating Outcomes	5
2.5.1	Distributions	5
2.6	Varying Initial Stat Count	5

1 Sub-Stat Probability

Unique combinations are computed and the probability of each corresponding permutation is calculated based on the stat weights.

The formula for the likelihood of a permutation, where w_i is the weight of sub-stat i , is:

$$\left(\frac{w_1}{Total}\right) \left(\frac{w_2}{Total - w_1}\right) \left(\frac{w_3}{Total - w_1 - w_2}\right) \cdots$$

For a given unique combination, all of these probabilities are added together for all permutations that correspond to that unique combination.

The sub-stat probability is then the sum of probabilities of combinations that are valid given the inputted constraints,

$$SubstatProbability = \sum_{valid} p$$

2 Roll Probability

The probability for a given sub-stat outcome is:

$$p \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

Where:

- p is the probability of the stat combination.
- $RollOutcomes$ is the number of possible roll outcomes.
- $InitialValueOutcomes$ the number of value outcomes for the initial states.
- $RollValueOutcomes$ the number of value outcomes for the rolled stats.

Since the sub-stat probability is separately defined above, we have:

$$RollProbability \times SubstatProbability =$$

$$p \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

$$RollProbability \times \sum_{valid} p =$$

$$p \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

$$RollProbability =$$

$$\left(\frac{p}{\sum_{valid} p}\right) \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

2.1 Total Number of Roll Possibilities

Let:

- a - The number of sub-stats
- a_r - The number of rollable sub-stats
- b - The number of roll values
- n - The number of sub-stat rolls

Then, there are $a = 4$ sub-stats to roll initially, $a^* = 4$ sub-stats choose from each time for each roll, and $b = 4$ roll values a roll can have (70%, 80%, 90%, and 100%).

- $InitialValueOutcomes = b^a = 4^4$ - There are 4 sub-stats to assign base values to
- $RollOutcomes = a_r^n = 4^n$ - There are n sub-stats rolls that each choose a sub-stat
- $RollValueOutcomes = b^n = 4^n$ - The n sub-stats rolls needs value assigned

Thus, the total number of possibilities is: $(b^{a+n})(a_r^n) = (4^{4+n})(4^n)$

2.2 Roll Permutations

To preserve the base of 4, each guaranteed roll is counted as $\frac{4}{\#of guaranteed rolls} = \frac{4}{2} = 2$.

To reduce computation, the computed 4^n roll permutations are combined into combinations, giving each effective outcome a value $RollPermutations$.

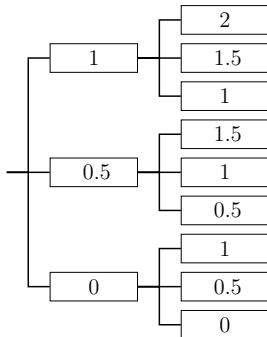
2.3 Initial and Roll Values

For each of these, the number of good outcomes is found based on the possible initial and roll values.

For n rolls, there are $4^{a+1}4^{b+1}4^{c+1}4^{d+1}$ possible initial and roll value combinations (or just $4^a4^b4^c4^d$ in the case of artifact re-rolling when the initial stats are preserved) where $a + b + c + d = n$.

A method computes the roll value distribution of $x = a + 1, \dots, d + 1$. It iterates x times, considering the 4 possible roll values that can extend the current computed values each time. To avoid computing a full 4^x outcomes, after each stage, paths with the same value are combined.

For example, assume the roll values are 0, 0.5, and 1. After the second iteration, we have:



Giving us $3^2 = 9$ outcomes to bring to the next iteration.

But instead, they can be grouped as so: 2 (1x), 1.5 (2x), 1 (3x), 0.5 (2x), 0 (1x). This leaves only 5 outcomes to bring to the next iteration. This is repeated for all of the x iterations.

The general method to compute the number of outcomes above the goal also uses this method, but for the 4 stat iterations.

This gives each outcome $InitialAndRollValuePossibilities_i$ for $i = 1, \dots, 4$.

2.4 Total Probability

The probability associated with a single outcome is now:

$$\frac{p \times RollPermutations \times \prod_{i=1}^4 InitialAndRollValuePossibilities_i}{\sum_{valid} p \times RollOutcomes \times InitialValueOutcomes \times RollValueOutcomes}$$

So, the total probability is:

TotalProbability

$$\begin{aligned} &= \sum_{combos}^{|stat|} \sum_{combos}^{|roll|} \sum_{combos}^{|value|} \frac{p \times RollPermutations \times \prod_{i=1}^4 InitialAndRollValuePossibilities_j}{\sum_{valid} p \times RollOutcomes \times InitialValueOutcomes \times RollValueOutcomes} \\ &= \sum_{combos}^{|stat|} \sum_{combos}^{|roll|} \sum_{combos}^{|value|} \frac{p}{\sum_{valid} p} \times \frac{RollPermutations}{RollOutcomes} \times \frac{\prod_{i=1}^4 InitialAndRollValuePossibilities_j}{InitialValueOutcomes \times RollValueOutcomes} \end{aligned}$$

2.5 Generating Outcomes

An overview of the code used to generate all outcomes is thus:

```

TotProb = 0
foreach ( $s_1, s_2, s_3, s_4 \rightarrow p$ ) of GenerateCombos do
  foreach ( $r_1, r_2, r_3, r_4 \rightarrow RollPermutations$ ) of RollCountsDist do
    States = Map( $0 \rightarrow 1$ )
    for  $i = 1 \dots 4$  do
      NextStates = Map()
      foreach ( $State \rightarrow Ways$ ) of States do
        foreach ( $statSum_i \rightarrow RollValuePossibilities_i$ ) of ValueOfRollsDist( $r_i + 1$ ) do
          NextState =  $State + statSum_i \times weight_i$ 
          NextStates[NextState] +=  $Ways \times RollValuePossibilities_i$ 
        end
      end
      States = NextStates
    end
    foreach ( $State \rightarrow Ways$ ) of States do
      if State is within goal then GoalWays += Ways
      TotWays += Ways
    end
    TotProb +=
      ( $GoalWays \div TotWays$ )  $\times$  ( $RollPermutations \div RollOutcomes$ )  $\times$  ( $p \div \sum_{valid} p$ )
  end
end
return TotProb

```

Note that *Ways* is just the product of the *RollValuePossibilities_i*, $i = 1, \dots, 4$, for the outcome.

2.5.1 Distributions

The distribution viewer provides a way to visualize *RollCountsDist* and *ValueOfRollsDist*.

2.6 Varying Initial Stat Count

Since artifacts have a chance to have 4 or 5 rolls (corresponding to 3 or 4 initially enabled stats), the actual probability is,

$$TotalProbability_{n=4} \times (1 - P(n = 5)) + TotalProbability_{n=5} \times P(n = 5)$$

So, the above pseudo-code is called twice for each calculation.